THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 9: **Light – Reflection and Refraction** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The radius of curvature of a concave mirror is 20 cm. Its focal length is: 1

(a) −20 cm **(b)** −40 cm **(c)** +10 cm **(d)** −10 cm

2. The power of a convex lens of focal length 50 cm is: 1

(a) +2 D **(b)** −2 D **(c)** +50 D **(d)** +0.5 D

3. The speed of light in a medium is 2×10^8 m/s. Its refractive index is ($c = 3 \times 10^8$ m/s): 1

(a) 1.5 **(b)** 0.67 **(c)** 2 **(d)** 1.33

4. A student states: “A convex mirror always forms a virtual, erect and diminished image.” The correct response is: 1

(a) Correct **(b)** Correct only for far objects

(c) Incorrect — it can form real images **(d)** Cannot be determined

5. For the mirror used as a rear-view mirror in vehicles, Student A says convex; Student B says concave. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. The image formed by a concave mirror is real, inverted and of the same size when the object is placed: 1

(a) at infinity **(b)** between F and P **(c)** at C **(d)** at F

7. A ray of light passing through the optical centre of a lens: 1

(a) goes undeviated **(b)** reflects back

(c) passes through the focus **(d)** becomes parallel to the axis

8. When light travels from air into glass, it bends: 1

(a) does not bend **(b)** away from the normal

(c) towards the normal **(d)** along the surface

9. The SI unit of power of a lens is: 1

(a) metre **(b)** watt **(c)** dioptre **(d)** joule

10. A lens produces a virtual, erect and magnified image. The lens is: 1

(a) plane glass **(b)** concave

(c) convex, object between F and O **(d)** convex, object beyond 2F

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State the two laws of reflection of light. 2

12. Define the principal focus of a concave mirror. 2

13. Why does a pencil appear bent when partly immersed in water? 2

14. A student says that a concave lens is used in a magnifying glass. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. An object is placed 20 cm in front of a concave mirror of focal length 15 cm. Find the position and nature of the image and the magnification. 3

16. State Snell's law of refraction. The refractive index of glass with respect to air is 1.5. What is the refractive index of air with respect to glass? 3

17. Draw ray diagrams (describe) to show the image formed by a concave mirror when the object is 3
(i) between F and C
(ii) between P and F.

OR

Write three uses of concave mirrors. 3

18. What is meant by the power of a lens? A concave lens has focal length 25 cm. Find its power. 3

OR

Define 3
(i) refractive index
(ii) optical centre
(iii) principal axis.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A 4 cm tall object is placed 15 cm from a convex lens of focal length 10 cm. 5
(i) Find the position, size and nature of the image.
(ii) Draw the ray diagram (describe).
(iii) What is the magnification?

OR

(i) Describe the image formation by a convex lens for the object at 5
(a) infinity
(b) beyond 2F
(c) at 2F.
(ii) State the lens formula.

20. **Case study:** A student traced the path of a ray through a rectangular glass slab and found the emergent ray parallel to the incident ray but shifted. 5
(i) Why is the emergent ray parallel?
(ii) What is the shift called?
(iii) What happens to speed of light in glass?
(iv) If $n = 1.5$, find the speed of light in glass.

OR

(i) State the New Cartesian sign convention for spherical mirrors. 5
(ii) Find the focal length of a convex mirror whose radius of curvature is 32 cm.
(iii) Why is it used in vehicles?

END OF QUESTION PAPER — Total: 40 Marks

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